K Ablations with the Alternative Format

We investigate the impact of prompting on several of the model families with highest amounts of overfitting. In this section, we test whether the difference in performance with the standard and alternative prompt is due to the alternative prompt using non-GSM8k examples. We do this by constructing prompts in the same format as the alternative "chain-of-thought" prompt but using fewshot example problems randomly chosen from GSM8k. These prompts imitate the alternative prompt's answer format and use of 8 fewshot examples rather than 5 in the standard prompt. We do this by converting two sets of randomly selected GSM8k problems into the analagous format.

We find significant variance in the results, ablating even something as simple as *which* n-shot examples are chosen. Nevertheless, the general shape of the findings remains largely consistent, even if the precise ordering / numerical values are highly prompt dependent.

For the first such ablation prompt (provided in full below), our results are displayed in Figure 14

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Q: Bob drove for one and a half hours at 60/mph.  He then hit
    construction and drove for 2 hours at 45/mph. How many miles did Bob
    travel in those 3 and a half hours?
A: Bob drove for 1.5 x 60 = 90 miles first, then another 2 x 45 = 90
    miles. In total Bob drove 90 + 90 = 180 miles. The answer is 180.

Q: Mary is paying her monthly garbage bill for a month with exactly four
    weeks. The garbage company charges Mary $10 per trash bin and $5 per
    recycling bin every week, and Mary has 2 trash bins and 1 recycling
    bin. They're giving her an 18%
A: Every week, Mary pays 10 x 2 = 20 dollars for the trash bins, so her
    total weekly cost is 20 + 5 = 25 dollars for the trash bins and the
    recycling bin. Then her monthly cost over 4 weeks is 25 x 4 = 100
    dollars. Mary's senior discount is 18 x .01 x 100 = 18 dollars. So
    subtracting the discount and adding the fine, her total monthly cost
    is 100 – 18 + 20 = 102 dollars. The answer is 102.

Q: June has $500 for buying school supplies for the new school year. She
    buys four maths books at $20 each, six more science books than maths
    books at $10 each, and twice as many art books as maths books at $20
    each. If she also bought music books, how much money did she spend on
     music books?
A: The total cost of maths books is 4 x 20 = 80. She bought six more
    science books than maths books which totals 6 + 4 = 10 books. If each
     science book cost her $10 she spent 10 x 10 = 100 dollars on science
     books. There were twice as many art books as maths books which total
     2 x 4 = 8. The total cost for art books is 8 x 20 = 160. The total
    amount that she used for maths, science, and art books is 160 + 100 +
     80 = 340. The amount she spent on music books is 500 – 340 = 160.
    The answer is 160.

Q: A play was held in an auditorium and its ticket costs $10. An
    auditorium has 20 rows and each row has 10 seats. If only 3/4 of the
    seats were sold, how much was earned from the play?
A: There are 20 x 10 = 200 seats in the auditorium. Only 200 x 3/4 = 150
    seats were sold. Hence, the earnings from the play is 10 x 150 =
    1500. The answer is 1500.

Q: Brendan went fishing with his dad. Brenden caught 8 fish in the
    morning. He threw 3 back that were too small. He caught 5 more in the
     afternoon. Brendan's dad caught 13 fish. How many fish did they
    catch in all?
A: Brenden caught 8 fish in the morning and 5 in the afternoon so he
    caught 8 + 5 = 13 total fish. After throwing the small fish back,
    Brenden has 13 – 3 = 10 fish. Together, Brenden and his dad caught 10
     + 13 = 23 fish. The answer is 23.
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Q: Valerie's cookie recipe makes 16 dozen cookies and calls for 4 pounds
    of butter.  She only wants to make 4 dozen cookies for the weekend.
    How many pounds of butter will she need?
A: Her original recipe makes 16 dozen and she only needs 4 dozen so she
    needs to reduce the recipe by 16 / 4 = 4. For 4 dozen cookies, she
    needs to reduce her recipe by 4 and the original called for 4 pounds
    of butter so she now needs 4 / 4 = 1 pound of butter. The answer is
    1.

Q: Jack is mad at his neighbors for blasting Taylor Swift all night, so
    he slashes three of their tires and smashes their front window. If
    the tires cost $250 each and the window costs $700, how much will
    Jack have to pay for the damages?
A: The total cost of the tires is 250 x 3 = 750. Then total cost of the
    tires and the window is 700 + 750 = 1450. The answer is 1450.

Q: A dental office gives away 2 toothbrushes to every patient who visits.
      His 8 hour days are packed and each visit takes .5 hours.  How many
     toothbrushes does he give in a 5 day work week?
A: Each day he does 8 / .5 = 16 visits. So he does 16 x 5 = 80 visits a
    week. That means he gives away 80 x 2 = 160 toothbrushes a week. The
    answer is 160.
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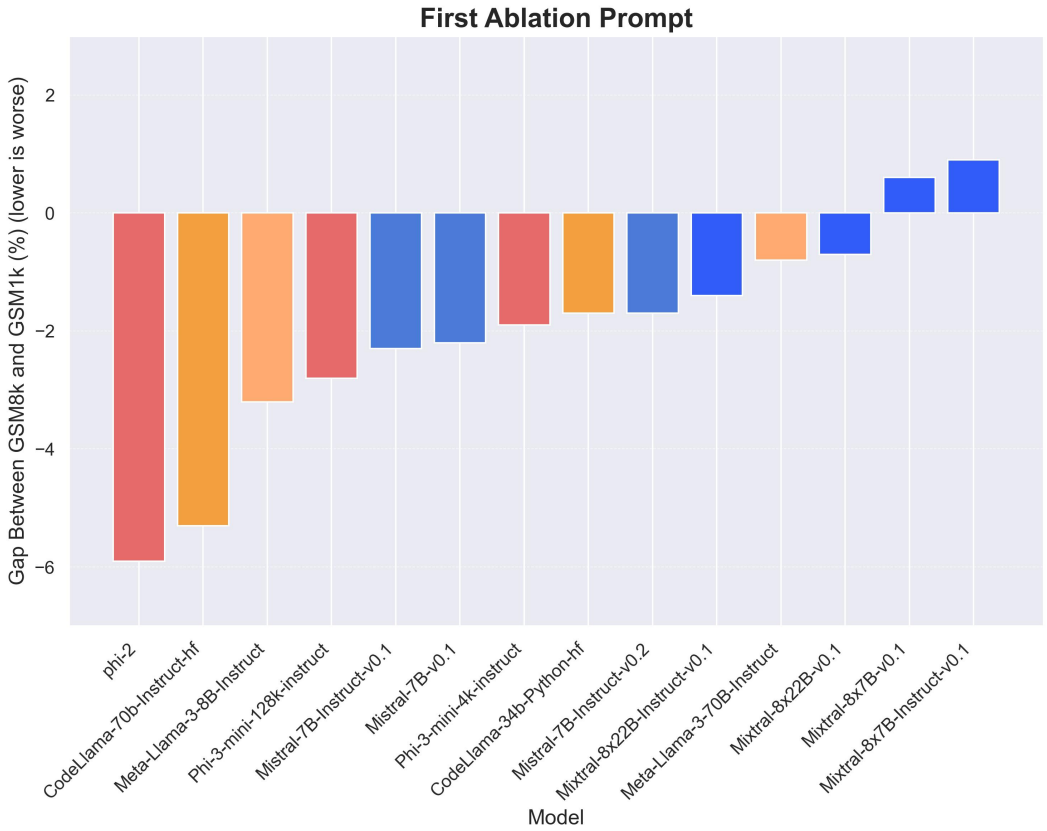

Figure 14: Models from the most overfit families arranged by their drop in performance between GSM8k and GSM1k (lower is worse) on the first ablation prompt with GSM8k examples in the alternative format.

For the second such ablation prompt (provided in full below), our results are displayed in Figure 15

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Q: At camp Wonka, there are 96 campers.  Two-thirds of the campers are
    boys, and the remaining one-third are girls.  50%
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A: The girls make up one—third of the campers, so there are 96 / 3 = 32 girls. The boys make up two—thirds of the campers, so there are 32 + 32 = 64 boys. There are 32 x 75%

Q: In today's field day challenge, the 4th graders were competing against the 5th graders. Each grade had 2 different classes. The first 4th grade class had 12 girls and 13 boys. The second 4th grade class had 15 girls and 11 boys. The first 5th grade class had 9 girls and 13 boys while the second 5th grade class had 10 girls and 11 boys. In total, how many more boys were competing than girls?
A: When you add up all the girls from all 4 classes, you had 12 + 15 + 9 + 10 = 46 girls. When you add up all the boys from all 4 classes, you had 13 + 11 + 13 + 11 = 48 boys. There are 48 boys and 36 girls so 48 — 46 = 2 more boys. The answer is 2.

Q: Axel bought an aquarium that was marked down 50%
A: The aquarium was bought for $120 x 50/100 = $60 less. So the marked down price of the aquarium was $120 — $60 = $60. Axel paid $60 x 5/100 = $3 additional for the sales tax. Therefore, the total cost of the aquarium was $60 + $3 = $63. The answer is 63.

Q: There are 48 crayons in the box. Kiley takes 1/4 of them away. Joe takes away half of the remaining crayons, how many crayons are left?
A: Kiley takes 48 / 4 = 12 crayons, so 48 — 12 = 36 crayons remain. Joe takes 36 / 2 = 18 crayons, so there are 36 — 18 = 18 crayons left. The answer is 18.

Q: Six Grade 4 sections launched a recycling drive where they collect old newspapers to recycle. Each section collected 280 kilos in two weeks . After the third week, they found that they need 320 kilos more to reach their target. How many kilos of the newspaper is their target?
A: In a week, each section collected 280 / 2 = 140 kilos of newspapers. So, in three weeks, one section collected 140 x 3 = 420 kilos. So, the four sections collected a total of 420 x 4 = 1680 kilos. Hence, their target is to collect 1680 + 320 = 2000 kilos of the newspaper. The answer is 2000.

Q: Jeff has a shelter where he currently takes care of 20 cats. On Monday he found 2 kittens in a box and took them to the shelter. On Tuesday he found 1 more cat with a leg injury. On Wednesday 3 people adopted 2 cats each. How many cats does Jeff currently have in his shelter?
A: Counting the cats he had, the kittens he found, and the injured cat, Jeff had a total of 20 + 2 + 1 = 23 cats. 3 people took a total of 3 x 2 = 6 cats. After Wednesday, Jeff was left with 23 — 6 = 17 cats. The answer is 17.

Q: Paul is working at a university. He is part of a big project, which involves 70 scientists in total. Half of them are from Europe and one —fifth are from Canada. The rest are from the USA. How many scientists in this project are from the USA?
A: Of all the scientists taking part in the project, half of them are from Europe, which means 70 x 0.5 = 35 people. The number of researchers from Canada is 70 x 1/5 = 14 people. That means there are 70 — 35 — 14 = 21 researchers from the USA. The answer is 21.

Q: John buys a heating pad for $30. He uses it 3 times a week for 2 weeks. How much does he spend on each use?
A: He uses it 3 x 2 = 6 times. So he pays $30 / 6 = $5 for each use. The answer is 5.